BayOmics msProteomiX -- Plasma Method Comparison User Manual

> Target audience: BayOmics marketing team
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1. Overview

This script compares different bead-based plasma proteomics sample preparation methods. It generates a **comprehensive evaluation report** with 14+ quality metrics, exported as a self-contained HTML file that can be printed to PDF.

What you get: A professional comparison report showing which method identifies more proteins, which has better quantification precision, depletion efficiency, etc.

2. Prerequisites

2.1 Install RStudio

Download and install RStudio Desktop from: https://posit.co/download/rstudio-desktop/

2.2 Install msProteomiX

Open RStudio, and run this in the Console (bottom-left panel):

```r
# Only needed once
install.packages("devtools")
devtools::install_github("BayOmics/msProteomiX")
install.packages("base64enc")   # For HTML report generation
```

2.3 Create a Project

In the RStudio Console, run:

```r
library(msProteomiX)
create_project("~/Desktop/PlasmaComparison")
```

This creates a project folder with all scripts and a `manuals/` folder.

3. Data Preparation

3.1 What to Request from the Bioinformatics Team

For each condition (e.g., "BayOmics_3uL", "Nanomics_5uL"), request:

Spectronaut results (3 files per condition):

File	Purpose	Example
`*_Report_ProtienGroup_MSC (Pivot).tsv`	Protein quantification	Required
`*_IdentificationsOverview.tsv`	Per-sample ID counts	Required
`*_Peptide_Report.tsv`	Peptide-level analysis	Optional but recommended

DIA-NN results (1 file per condition):

File	Purpose
`*_report.pg_matrix.tsv`	Protein quantification

3.2 Organize Files

Place each condition's files in its own subfolder under `wkdir/` :

```
PlasmaComparison/
  wkdir/
    Spectronaut/
      Condition_A/        <-- Put all 3 files here
      Condition_B/        <-- Put all 3 files here
    DIA_NN/
      pg_matrix_A.tsv     <-- One file per condition
      pg_matrix_B.tsv
  scripts/
    19_...R               <-- The script you will edit
  output/                 <-- Results will appear here
```

4. Edit the Configuration

Open `scripts/19_血浆前处理方案比较.R` in RStudio.

You only need to edit the **CONFIG** section (lines 22-42). Everything below the line `# EXECUTION -- Do NOT edit below this line` should not be touched.

4.1 Set Conditions

Edit the `conditions` list. Each condition needs:

- `name` : A short label (use English, no spaces)
- `path` : Relative path to the data folder (starting with `./wkdir/`)
- `engine` : `"spectronaut"` or `"diann"`

Example (comparing 2 methods, each with 2 volume conditions):

```
conditions <- list(
  list(name = "BayOmics_3",    path = "./wkdir/Spectronaut/BM_3",    engine = "spectronaut"),
  list(name = "BayOmics_5",    path = "./wkdir/Spectronaut/BM_5",    engine = "spectronaut"),
  list(name = "Competitor_3", path = "./wkdir/Spectronaut/NM_3",    engine = "spectronaut"),
  list(name = "Competitor_5", path = "./wkdir/Spectronaut/NM_5",    engine = "spectronaut")
)
```

4.2 Set Groups

Edit `compare_groups` to define which conditions belong to which group.
Groups determine the color coding in charts.

```r
compare_groups <- list(
  BayOmics   = c("BayOmics_3", "BayOmics_5"),
  Competitor = c("Competitor_3", "Competitor_5")
)
```

4.3 Set Project Name

```r
project_name <- "Customer_PlasmaG"   # Appears in report header
```

5. Run the Script

1. In RStudio, open the script file
2. Press **Ctrl+A** (select all)
3. Press **Ctrl+Enter** (run)
4. Wait for the message: `All evaluations complete!`

This typically takes 1-3 minutes depending on the number of conditions.

6. Find Your Results

After the script finishes, check the `output/` directory:

```
output/
  BayOmics_3_evaluation/      <-- Per-condition CI + QC results
  Nanomics_3_evaluation/
  ...
  comparison/                  <-- Cross-condition comparison
    10_comparison_report_en.html   <-- English report (main deliverable)
    10_comparison_report_en.pdf    <-- English PDF (if Chrome installed)
    10_comparison_report_zh.html   <-- Chinese report
    10_comparison_report_zh.pdf    <-- Chinese PDF
    10_pg_comparison.pdf           <-- Individual charts (for PPT use)
    11_precursor.pdf
    ...
```

6.1 The Main Report

Open `10_comparison_report_en.html` (or `_zh.html`) in your web browser.
This is a **self-contained** file with all charts embedded. You can:

- **Email it directly** to clients
- **Print to PDF**: File > Print > Save as PDF (if auto-PDF failed)

6.2 Individual Charts

All comparison charts are also saved as individual PDF files in `output/comparison/` for use in PowerPoint presentations.

7. Troubleshooting

Error	Solution
`Error: package 'msProteomiX' not found`	Run `devtools::install_github("BayOmics/msProteomiX")`
`Error: package 'base64enc' not found`	Run `install.packages("base64enc")`
`Error: cannot open file '...'`	Check that the file path is correct and the file exists
`Chrome not found`	Install Google Chrome, or open the HTML file in a browser and use Print > Save as PDF
Chinese characters in path cause errors	Move data to a path with only English characters

8. Appendix: What Each Metric Means

Metric	What It Measures	Good Result
Protein Groups	How many proteins were identified	More is better
Precursors	How many peptide precursors identified	More is better
Data Completeness	% proteins detected in all replicates	Higher is better
Quant CV	Reproducibility of protein quantities	Lower is better (<20% ideal)

Metric	What It Measures	Good Result
Correlation	Agreement between replicates	Higher is better (>0.95 ideal)
PCA	Sample clustering pattern	Replicates should cluster together
CI (Contamination Index)	Blood cell contamination level	Pass = clean sample
Top-N Fraction	How much signal comes from top proteins	Lower = better depletion
High-Abundance Rank	Where ALB/IgG rank in intensity	Higher rank = better depletion
Missed Cleavage	Digestion efficiency (0-miss rate)	Higher 0-miss = better digestion
Peptide Length	Length of identified peptides	7-25 aa is typical
Dynamic Range	Intensity span (P1-P99)	Wider = better sensitivity
Rank Curve	Protein abundance distribution	Smooth curve is expected

Manual by BayOmics Bioinformatics Team. For support, contact the bioinformatics team.